

Ahmad Reza Abdollahi

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EDUCATION:

Penn State Erie, The Behrend College

Bachelor of Science in Software Engineering

Graduation: May 2027

GPA: 3.99

RESEARCH EXPERIENCE:

Undergraduate Research Assistant | Penn State Erie

Spring 2024

- Focus on learning about CNN models and specifically YOLO architecture for real object detection and classification with high accuracy.
- Review seminal works, including AlexNet, to gain a comprehensive understanding of convolutional layers and their functionalities.

Undergraduate Research Assistant | Penn State Erie

Summer 2024

- Develop a prototype dataset by scraping text from the university's website, designed for fine-tuning text-to-text and text generation models.
- Create a complementary dataset tailored for question-answering models.
- Fine-tune various small-scale Large Language Models (LLMs), including T5-based and Llama-3.1-8B.
- Formulated a framework to assist developers in selecting appropriate LLMs based on project-specific requirements.

PERSONAL PROJECTS:

- Utilize ResNet101 to develop a deep learning model for classifying images in the Food101 dataset.
- Design and developed a custom deep learning model for classifying images in FashionEMNIST and EMNIST datasets, enhancing classification accuracy and model performance.
- Develop and implement a predictive model for the Ames Housing dataset, focusing on accurate property value estimation and feature optimization.
- Fine-tune large language models (LLMs) on a university-specific dataset to enhance their performance in answering academic queries and maintaining conversational interactions.
- Develop a website that leverages natural language processing to summarize texts, providing users with concise and relevant information from lengthy content.

ADDITIONAL WORK EXPERIENCE:

**Tutor of Math and science, Introduction to Programming Techniques,
Intermediate Programming** | Penn State Erie

Spring 2024 - Present

Guitar Teacher | Iran

Fall 2019 - Present

Lifeguard | YMCA

Summer2024 - Present

Front Desk | LRC of Penn State Behrend

Fall 2024 - Present

Technical Skills:

- **Programming languages:** C++, Python, Java, JS, HTML, CSS, VHDL, SQL
- **Frameworks:** React.JS, PyTorch, TensorFlow
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Technologies:** Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP)
- **Natural language:** English (fluent), Persian (fluent)
- **Development Tools:** Visual Studio, VS code, IntelliJ, Eclipse, Git